

We need to show that:

$$\sum_{i=1}^m \sum_{j=1}^n (x_i + y_j) = n \sum_{i=1}^m x_i + m \sum_{j=1}^n y_j$$

Here is the step-by-step proof.

Step 1: Expand the inner sum (with respect to j)

We start with the left-hand side (LHS):

$$\sum_{i=1}^m \left[\sum_{j=1}^n (x_i + y_j) \right]$$

Let's focus on the inner summation $\sum_{j=1}^n (x_i + y_j)$. For a fixed value of i , we are summing $x_i + y_j$ as j goes from 1 to n :

$$\sum_{j=1}^n (x_i + y_j) = (x_i + y_1) + (x_i + y_2) + \cdots + (x_i + y_n)$$

Notice that x_i appears exactly n times. So we can group the x_i terms and the y_j terms:

$$\begin{aligned} &= \underbrace{x_i + x_i + \cdots + x_i}_{n \text{ times}} + (y_1 + y_2 + \cdots + y_n) \\ &= n \cdot x_i + \sum_{j=1}^n y_j \end{aligned}$$

Step 2: Substitute the expanded inner sum back into the outer sum

Now we replace the inner sum in our original LHS expression with the result from Step 1:

$$\text{LHS} = \sum_{i=1}^m \left(n \cdot x_i + \sum_{j=1}^n y_j \right)$$

Step 3: Distribute the outer sum (with respect to i)

Using the distributive property of summation, $\sum(A + B) = \sum A + \sum B$, we get:

$$\text{LHS} = \sum_{i=1}^m (n \cdot x_i) + \sum_{i=1}^m \left(\sum_{j=1}^n y_j \right)$$

Step 4: Simplify each term

First term: Since n is a constant, it can be factored out of the summation:

$$\sum_{i=1}^m (n \cdot x_i) = n \sum_{i=1}^m x_i$$

Second term: The expression $\sum_{j=1}^n y_j$ is a constant with respect to the index i . We are simply adding this same constant value m times (as i goes from 1 to m):

$$\sum_{i=1}^m \left(\sum_{j=1}^n y_j \right) = m \cdot \left(\sum_{j=1}^n y_j \right)$$

Step 5: Combine to reach the Right-Hand Side (RHS)

Putting the simplified terms together:

$$\text{LHS} = n \sum_{i=1}^m x_i + m \sum_{j=1}^n y_j$$

This matches the right-hand side of equation exactly:

$$\text{RHS} = n \sum_{i=1}^m x_i + m \sum_{j=1}^n y_j$$

$$\boxed{\therefore \sum_{i=1}^m \sum_{j=1}^n (x_i + y_j) = n \sum_{i=1}^m x_i + m \sum_{j=1}^n y_j}$$